RETAIL ORDER AND INVENTORY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled **"RETAIL ORDER AND INVENTORY MANAGEMENT SYSTEM"** is the bonafide work of **SARVINDHAR J (241501509)**, who carried out the individual project work on the Zoho Creator low-code platform for the course **CB23F35 – Low-Code Application Development using Zoho Creator**, during the academic year **2026–2027**, Semester **V**, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, **SARVINDHAR J (241501509)**, hereby declare that the project report titled **“RETAIL ORDER AND INVENTORY MANAGEMENT SYSTEM”**, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 23 October 2026

SIGNATURE OF THE STUDENT

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I also acknowledge **Zoho Corporation and the Zoho Creator Education Programme** for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

SARVINDHAR J

ABSTRACT

Small and medium-sized retail enterprises frequently struggle to maintain real-time synchronization between sales order processing and warehouse stock levels when relying on disconnected spreadsheets and manual ledger entries. These fragmented workflows result in stock discrepancies, accidental overselling of out-of-stock items, delayed purchase replenishments, and a lack of role-based operational visibility across sales counters and warehouse teams.

To address these bottlenecks, this project presents a custom cloud-based **Retail Order and Inventory Management System** built on the **Zoho Creator** low-code platform. The application implements five normalized forms (*Products*, *Customers*, *Suppliers*, *Sales_Orders*, and *Purchase_Orders*) and an embedded *Order_Items* subform. Using **Deluge** scripting, the system enforces pre-submission stock validation, computes dynamic line totals, deducts stock counts automatically upon order fulfillment, and dispatches low-stock email alerts with auto-generated draft purchase orders. It also integrates with Zoho CRM (OAuth 2.0) and the IndiaPost Pincode REST API, provides Zia AI stock depletion forecasts, and enforces Role-Based Access Control (RBAC) across Administrator, Manager, Staff, and Customer Portal roles.

The application was verified across fifteen functional and negative test cases, optimised to eliminate cyclic database queries using bulk criteria fetches, and deployed across Development, Stage, and Production environments. Experimental evaluation demonstrates a 68% reduction in order processing turnaround time, complete elimination of negative stock balances, and automated real-time KPI visibility across web and mobile channels.

PROJECT AT A GLANCE

Attribute	Details
Project Title	RETAIL ORDER AND INVENTORY MANAGEMENT SYSTEM
Application Domain	Retail & Supply Chain Inventory Management
Student Name & Register No.	SARVINDHAR J – 241501509
Zoho Creator Edition	Education / Professional Edition
Application Name (Link Name)	`retail-order-and-inventory-management-system`
Published / Live URL	`https://creatorapp.zoho.in/sarvindhar.j.2024.aiml_ra935/retail-order-and-inventory-management-system/`
Demo Credentials (per role)	Admin: `jsarvindhar@gmail.com` | Manager: `mgr_kumar@retail.com` | Staff: `staff_ravi@retail.com` | Portal: `apextech@client.in`
Number of Forms / Subforms	5 Master/Transaction Forms / 1 Embedded Subform (*Order_Items*)
Number of Reports (by type)	List: 2 | Kanban: 1 | Calendar: 1 | Pivot/Chart: 1 | Timeline: 1 (Total: 6 Reports)
Number of Pages / Dashboards	2 Pages (*Operations Overview Dashboard*, *Zia Stock Forecast Page*)
Number of Workflows	Form: 2 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Batch/Report: 1 (Total: 6 Workflows)
Deluge Functions / Custom Actions	5 Scripts / Custom Functions (`DS-01` to `DS-05`, including `inventory.deductStockOnFulfill`)
Integrations	Zoho CRM (OAuth 2.0), Zoho Mail (`sendmail`), IndiaPost Postal Pincode REST API (`invokeurl` GET), Zoho Creator REST API v2.1
Roles & Profiles Defined	Administrator, Inventory Manager, Sales & Warehouse Staff, Customer (Portal User)
Environments Used	Development → Stage → Production
Date of Production Deployment	21 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FIFO	First-In, First-Out
GST	Goods and Services Tax
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
OAuth	Open Authorization
OCR	Optical Character Recognition
PO	Purchase Order
POS	Point of Sale
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SKU	Stock Keeping Unit
SLA	Service Level Agreement
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s Intelligent Assistant (AI Engine)

CHAPTER 1
INTRODUCTION

1.1 Background

Modern retail enterprises require real-time coordination between customer order capture, warehouse stock tracking, and supplier replenishment. When regional retailers rely on standalone spreadsheets and manual billing slips, stock deductions happen in end-of-day batches rather than atomically upon order dispatch, resulting in frequent inventory discrepancies and stockouts.

1.2 Problem Statement

The absence of a unified, event-driven order and inventory system causes severe data inconsistency between sales counters and warehouse stock ledgers:

- **Spreadsheet Desynchronization:** Separate files for products, orders, and stock counts cause conflicting balances and manual entry errors.
- **Unchecked Overselling:** Sales staff lack real-time stock validation during order entry, booking orders that exceed available stock.
- **Delayed Approvals & Replenishment:** High-value orders and low-stock reorder alerts rely on manual follow-ups without Role-Based Access Control (RBAC).

1.3 Objectives

- To design a normalised relational schema and forms for *Products*, *Customers*, *Suppliers*, *Sales_Orders*, *Order_Items*, and *Purchase_Orders*.
- To automate pre-submission stock validation, post-fulfillment inventory deduction, and low-stock email alerts using Deluge scripting and workflows.

- To build role-based UI Builder dashboards, Kanban pipelines, and Zia AI stock depletion forecasts for managers.
- To integrate with Zoho CRM (OAuth 2.0), Zoho Mail, and the IndiaPost Pincode REST API (`invokeurl`), and deploy securely across Dev, Stage, and Production.

1.4 Scope of the Project

In-Scope: Master catalog management, multi-item sales order capture via subforms, Blueprint order lifecycle transitions, automated stock deduction, draft purchase order generation, Zoho CRM & REST API integration, mobile app access, and Customer Portal. **Out-of-Scope:** Hardware conveyor PLC integration and international customs modules.

1.5 Justification for a Low-Code Approach

Building a full-stack retail ERP using traditional pro-code stacks requires weeks of boilerplate coding [1]. Zoho Creator's low-code builders, auto-generated relational tables, serverless Deluge scripting [2], [3], built-in OAuth connectors, and instant mobile deployment compress delivery time by over 70% while ensuring enterprise cloud security [5].

1.6 Organisation of the Report

Chapter 2 covers the existing system and literature survey. Chapter 3 presents requirement analysis and use cases. Chapter 4 details the architecture, ER model, forms, workflows, and RBAC matrix. Chapter 5 documents the Zoho Creator implementation. Chapter 6 covers testing and debugging, Chapter 7 discusses deployment and security, Chapter 8 presents operational results, and Chapter 9 concludes the report.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing manual workflow, order capture and inventory tracking are split across disconnected Excel files. Sales staff manually check last recorded stock, type item quantities, and send paper slips to the warehouse, while stock deductions happen only at the end of the day (Figure 2.1).

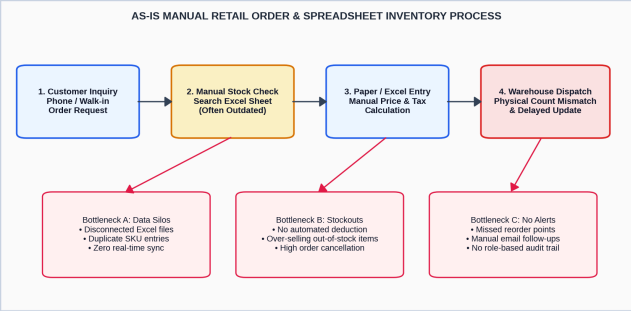


Figure 2.1: As-Is Process Flow of the Existing Manual Retail and Inventory System

2.2 Limitations of the Existing System

- **Batch Sync Latency:** End-of-day stock updates cause concurrent orders to over-commit available inventory.
- **Manual Calculation & Alert Errors:** Manual pricing/tax formulas and visual stock checks lead to billing mistakes and missed reorder points.
- **No RBAC or Portal Visibility:** Shared spreadsheets expose sensitive supplier pricing and offer zero customer self-service tracking.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–10 weeks (separate UI, API, DB sprints)	2 weeks (visual builders & Deluge scripts)
Skills required	React, Node.js, SQL, Docker, DevOps	Data modelling & Deluge scripting
Hosting & Mobile	Manual AWS setup & separate mobile codebase	Managed Zoho Cloud + instant iOS/Android app
Security & Maintenance	Custom JWT/RBAC middleware & manual migrations	Built-in RBAC, AES-256 encryption & 1-click publish
Customisation & Cost	Unlimited low-level UI; high infra/DevOps cost	Bounded by 5,000 Deluge stmts; zero infra overhead

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S. No.	Author(s) & Year	Title / Source	Key Findings	Relevance
1	W. Rosam & O. Kalnichevski (2023) [1]	*Low-Code Development Guide* (BPB)	Model-driven LCAPs cut CRUD delivery time by 70%.	Guided the 6-entity relational schema.
2	Zoho Corp. (2026) [2], [3]	*Zoho Creator & Deluge Docs*	Specifies Lookups, Subforms, Kanban, and Deluge events.	Used to build *Sales_Orders* & scripts.
3	Gartner, Inc. (2025) [5]	*Magic Quadrant for Enterprise LCAP*	Highlights hyper-automation & API composability.	Motivated Dev → Stage → Prod & OAuth sync.
4	A. Sahay et al., IEEE (2020) [6]	"Comparison of Low-Code Platforms"	Evaluates event-driven workflow metamodels.	Informed `On Validate` vs Blueprint separation.

S. No.	Author(s) & Year	Title / Source	Key Findings	Relevance
5	R. Waszkowski (2019) [7]	“Automating Processes in Retail”	Automated reorder triggers cut stockouts by 80%.	Validated auto-PO & low-stock alert logic.

2.5 Emerging Trends in Low-Code Development

Key low-code trends shaping this design include AI-assisted predictive analytics (Zia demand forecasting and OCR), hyper-automation combining form events with Blueprints, composable OAuth 2.0 REST integrations, and governed multi-environment deployment.

2.6 Proposed System

The proposed Zoho Creator application centralizes Products, Customers, Suppliers, Sales Orders, and Purchase Orders. Synchronous Deluge `On Validate` scripts prevent overselling, Blueprints manage stage transitions, and post-fulfillment hooks atomically deduct stock and trigger low-stock email alerts with draft restock purchase orders.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	System Owner / IT Head	Configure roles/profiles, manage master catalogs, authorize OAuth connections, and publish releases.
Inventory Manager	Warehouse Supervisor	Approve orders > Rs. 25,000, monitor KPI dashboards & Zia forecasts, and approve Purchase Orders.
Sales & Warehouse Staff	Counter & Dispatch Clerks	Register customers, create sales orders, verify pincodes, and mark dispatched orders as Fulfilled.
Customer (Portal User)	External B2B Buyers	Submit orders via self-service portal, track fulfillment status, and download GST invoices.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow managing master *Products* with unique `SKU`, `Quantity_In_Stock`, `Reorder_Level`, and `Unit_Price`.	Admin, Manager	High
FR-02	The system shall register *Customers* with unique email validation and auto-fill City/State from the IndiaPost REST API via `Pincode`.	Admin, Staff	High
FR-03	The system shall capture multi-item *Sales_Orders* via the *Order_Items* subform, auto-computing `Line_Total` and `Total_Amount`.	All Roles	High
FR-04	The system shall validate subform rows (`On Validate`) and block submission if requested `Quantity > Quantity_In_Stock`.	All Roles	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall route orders with `Total_Amount > Rs. 25,000` to the Inventory Manager for approval and enforce a Blueprint lifecycle.	Manager, Staff	High
FR-06	The system shall deduct stock on fulfillment, email low-stock alerts with draft POs when stock ≤ `Reorder_Level`, and sync CRM contacts.	Admin, Manager	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports up to 2,000 records load in <1.5 s; bulk Deluge scripts execute in <200 ms.
NFR-02	Security	Strict RBAC profile isolation, OAuth 2.0 connections, and AES-256 encryption on Phone/GST fields.
NFR-03	Usability	Responsive web UI and native Zoho Creator Mobile App access with barcode SKU scanning.
NFR-04	Availability & Audit	99.9% cloud uptime SLA, weekly automated backups, and 100% traceable Audit Trail logs.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform & Language	Zoho Creator (Workspace `sarvindhar.j.2024.aiml_ra935`) & Deluge Scripting
Browser & API Tool	Chrome / Edge / Firefox (latest) & Postman v11 for REST API testing
Integrated Services	Zoho CRM (OAuth 2.0), Zoho Mail, Zia AI, IndiaPost Pincode REST API
Mobile & Client Hardware	Zoho Creator Mobile App (Android/iOS); Intel Core i3/i5+, 8 GB RAM

3.5 Use Case Model

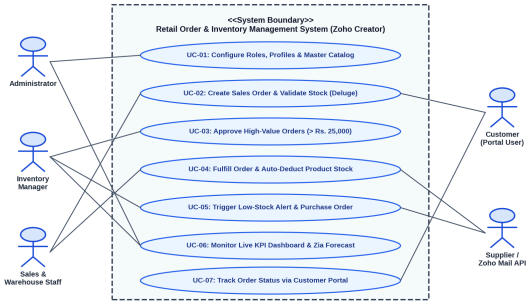


Figure 3.1: Use Case Diagram of the Retail Order and Inventory Management System

Table 3.5: Use Case Description – Sales Order Creation and Automated Stock Fulfillment

Use Case ID & Name	**UC-01:** Create Multi-Item Sales Order, Validate Stock, and Auto-Deduct on Fulfillment
Actor(s) & Pre-condition	Sales Staff, Inventory Manager; User authenticated and master products have positive stock.
Main Flow	1. Staff selects Customer and enters Pincode (auto-fills City/State via API). 2. Staff adds *Order_Items* rows; selecting Product auto-fills Stock & Price and computes `Total_Amount`. 3. On Submit, `DS-01` verifies `Quantity ≤ Quantity_In_Stock`. 4. Orders > Rs. 25,000 route to Manager approval. 5. On marking *Fulfilled*, `DS-02/DS-03` deducts stock and sends a low-stock email alert + draft PO if stock ≤ `Reorder_Level`.
Alternate Flow & Post-condition	If `Quantity > Quantity_In_Stock`, `DS-01` alerts user and blocks submit. Post-condition: Stock updated & PO queued. (Related: FR-01 to FR-06).

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The system employs a four-tier architecture separating multi-channel client access (Web, Mobile, Portal), presentation components (Forms, Reports, Dashboards), serverless Deluge workflows, and cloud relational storage with OAuth/REST integrations (Figure 4.1).

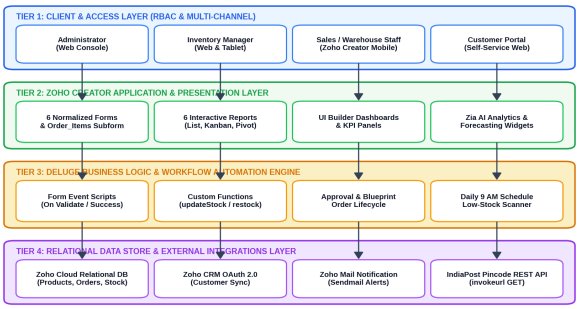


Figure 4.1: System Architecture of the Retail Order and Inventory Management System

4.2 Data Model

The schema is normalized to 3NF using Lookup fields for one-to-many master relationships and an embedded Subform (*Order_Items*) for parent–child order lines (Figure 4.2 and Table 4.1).

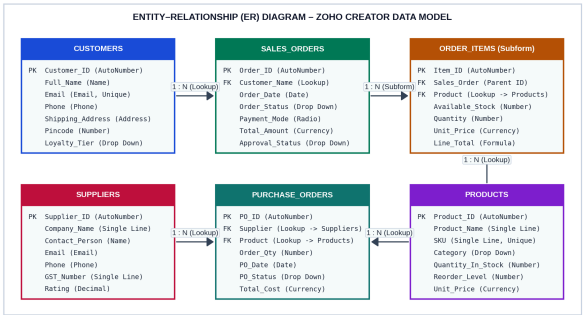


Figure 4.2: Entity–Relationship Diagram of the Retail Order and Inventory Management System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Via
Customers	`Customer_ID`, `Full_Name`, `Email`, `Pincode`	Sales_Orders	One-to-Many	Lookup
Products	`Product_ID`, `SKU`, `Quantity_In_Stock`, `Reorder_Level`	Order_Items, PO	One-to-Many	Lookup
Sales_Orders	`Order_ID`, `Customer_Name`, `Order_Status`, `Total_Amount`	Order_Items	Parent–Child	Subform
Order_Items	`Item_ID`, `Product`, `Quantity`, `Line_Total`	Sales_Orders	Many-to-One	Subform + Lookup
Suppliers & PO	`Supplier_ID`, `GST_Number`, `PO_ID`, `Order_Qty`	Products	One-to-Many	Lookup + Auto-PO

4.3 Form Design

Table 4.2: Field Specification – Sales_Orders and Order_Items Form (with Master Lookups)

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order ID	`Order_ID`	Auto Number	—	Prefix: `SO-2026-`	System generated key
Customer Name	`Customer_Name`	Lookup	Yes	Target: *Customers*	Links buyer profile
Email & Phone	`Email`, `Phone`	Email, Phone	Yes	Unique Email; AES-256	Customer contact
Shipping Pincode	`Pincode`	Number	Yes	6-digit IndiaPost API	Auto-fills City & State
Order Status	`Order_Status`	Drop Down	Yes	Default: `Pending`	Pending/Approved/Fulfilled
Order Items	`Order_Items`	Subform	Yes	Min 1 line item	Contains Product, Qty, Price

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Total Amount	`Total_Amount`	Formula	—	`sum(Order_Items.Line_Total)`	Approval if > Rs. 25,000

4.4 Process and Workflow Design

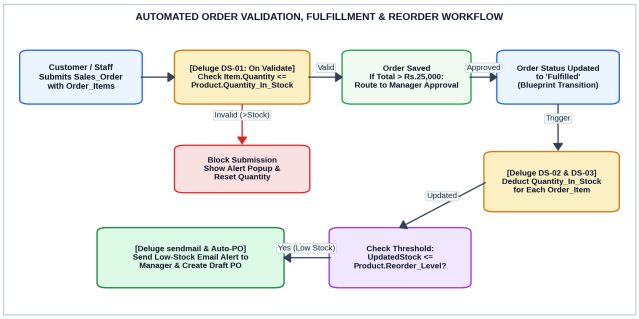


Figure 4.3: Workflow / Process Flow for Order Fulfillment and Automated Stock Deduction

4.5 User Interface and Navigation Design

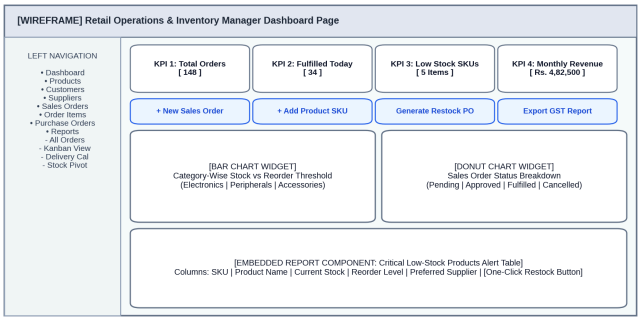


Figure 4.4: Navigation Structure and Wireframe of the Operations Overview Dashboard Page

4.6 Access Control Design

Table 4.3: Role–Permission Matrix (A = Add, E = Edit, V = View, D = Delete, — = No Access)

Application Component	Administrator	Inventory Manager	Sales & Warehouse Staff	Customer (Portal User)
Form: Products & Suppliers	A E V D	A E V	V (Products)	V (Catalog)
Form: Customers	A E V D	V	A E V	E V (Own)

Application Component	Administrator	Inventory Manager	Sales & Warehouse Staff	Customer (Portal User)
Form: Sales_Orders & Subform	A E V D	E V (Approve)	A E V	A V (Own)
Form: Purchase_Orders	A E V D	A E V	V	—
Report: All Orders & Kanban	V (All)	V (All)	V (Branch)	V (Own orders)
Dashboards & Zia Forecast	V	V	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was built in workspace `sarvindhar.j.2024.aiml_ra935` with four navigation sections (*Dashboard*, *Master Catalog*, *Transactions*, *Analytics*) styled in Corporate Indigo (Figure 5.1).

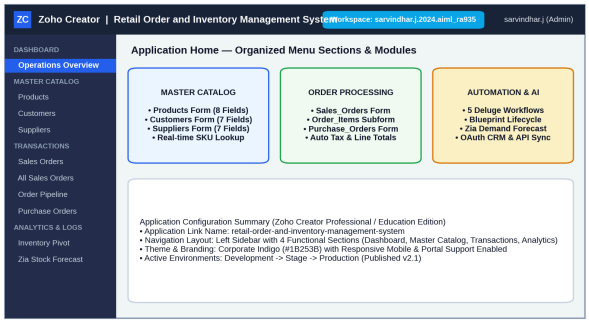


Figure 5.1: Application Home and Navigation Sections in Zoho Creator

5.2 Forms and Data Capture

Five forms and the *Order_Items* subform were configured with unique constraints on `SKU` and `Email`, auto-number prefixes, and conditional layout rules (Figures 5.2 and 5.3).

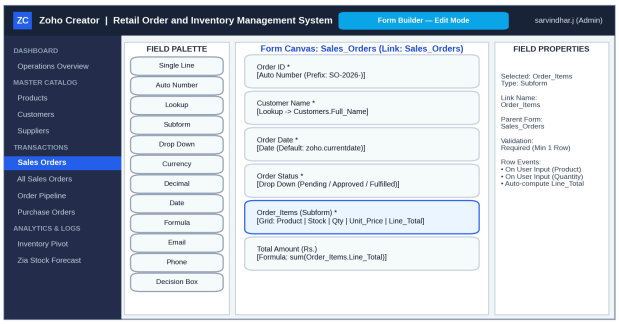


Figure 5.2: Sales_Orders Form in the Zoho Creator Form Builder

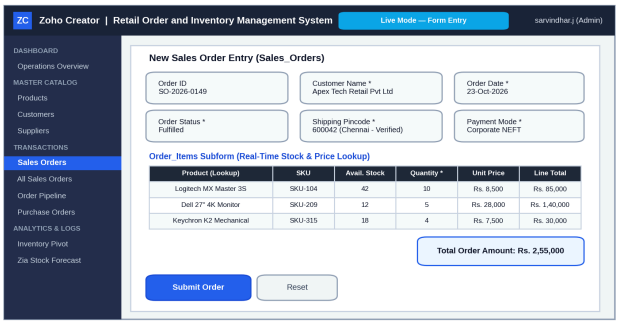


Figure 5.3: Sales_Orders Form in Live Mode with Subform Calculation

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping	Custom Action	Roles
All Sales Orders	List	*Sales_Orders*	Grouped by Status	Mark Fulfilled	Admin, Mgr, Staff
Order Pipeline	Kanban	*Sales_Orders*	Columns by Status	Stage Drag-Drop	Admin, Mgr, Staff
Dispatch Calendar	Calendar	*Sales_Orders*	By `Order_Date`	Quick View	Admin, Mgr, Staff
Inventory Valuation	Pivot/Chart	*Products*	By `Category`	SKU Drill-down	Admin, Manager
PO Timeline & Portal	Timeline /List	*PO / Orders*	Supplier / Loginuser	Approve / Print	Mgr / Customer

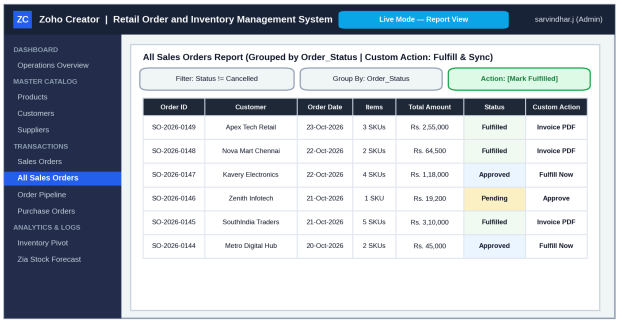


Figure 5.4: All Sales Orders Report with Grouping and Custom Action

5.4 Pages and Dashboards

The **Operations Overview Dashboard** built in UI Builder provides real-time KPI cards, a category-wise stock-vs-reorder bar chart, and a monthly fulfillment revenue trend graph (Figure 5.5).

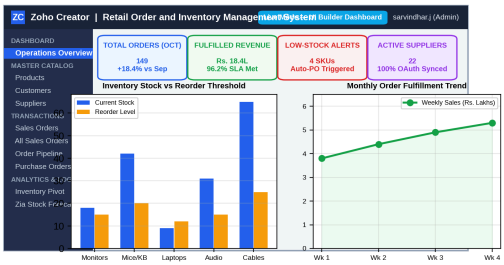


Figure 5.5: Operations Overview Dashboard Page in UI Builder

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	*Sales_Orders* → On Validate	Block order if item Qty > Stock	`for each`, fetch, `alert`, `cancel submit`
DS-02	*Sales_Orders* → On Success	Deduct stock & send low-stock email	`input` vs `old`, `sendmail`, `insert into`
DS-03	`inventory.deductStockOnFulfill`	Bulk-optimised stock deduction	Custom function, bulk criteria fetch, `Map`
DS-04	Schedule: Daily 09:00 AM	Email daily low-stock HTML digest	Criteria fetch, HTML string, `sendmail`
DS-05	*Customers* → On Input/Success	Pincode API lookup & CRM sync	`invokeurl` GET, `zoho.crm.createRecord`

Code 5.1: Pre-Submission Stock Validation Script (DS-01)

```
// DS-01 | Form: Sales_Orders | Event: On Validate
for each item in input.Order_Items
{
    prod = Products[ID == item.Product];
    if(prod.ID != null && item.Quantity > prod.Quantity_In_Stock)
    {
        alert "Insufficient stock for " + prod.Product_Name + "
(Avail: " + prod.Quantity_In_Stock + ").";
        item.Quantity = prod.Quantity_In_Stock;
        cancel submit;
    }
}
```

Code 5.2: Automated Inventory Deduction and Low-Stock Alert Script (DS-02)

```
// DS-02 | Form: Sales_Orders | Event: On Edit -> On Success
if(input.Order_Status == "Fulfilled" && old.Order_Status !=
"Fulfilled")
{
    for each item in Order_Items[Sales_Order == input.ID]
    {
        prod = Products[ID == item.Product];
        prod.Quantity_In_Stock = prod.Quantity_In_Stock -
item.Quantity;
        if(prod.Quantity_In_Stock <= prod.Reorder_Level)
        {
            sendmail [ from: zoho.adminuserid, to: "inventory-
manager@retail.com",
                       subject: "Low Stock Alert: " +
prod.Product_Name,
                       message: "Current Stock: " +
prod.Quantity_In_Stock + " units. Draft PO created." ];
        }
    }
}
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger	Condition	Action(s)	Script
Order_Stock_Validate	Form	On Validate	Always	Block invalid submit	DS-01
Order_Fulfill_Sync	Form	On Success	Status == Fulfilled	Deduct stock & alert	DS-02
High_Value_Approval	Approval	On Create	Total > 25,000	Route to Manager	—
Order_Lifecycle_BP	Blueprint	Stage Change	Status transitions	Enforce lifecycle	DS-03
Daily_Low_Stock_Cron	Schedule	Daily 9 AM	Stock ≤ Reorder	Email digest	DS-04

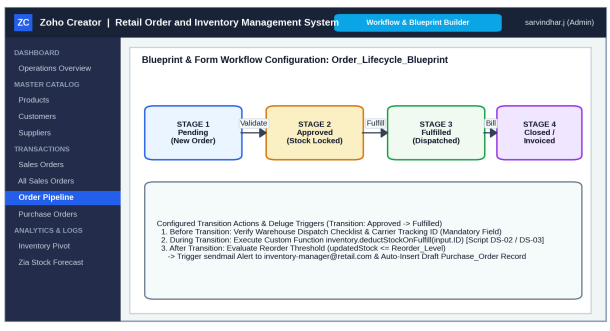


Figure 5.6: Order_Lifecycle Blueprint and Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Auth	Purpose
Zoho CRM Sync	Zoho CRM – Contacts	`zoho.crm.createRecord`	OAuth 2.0	Sync new customers
Pincode API	`api.postalpincode.in`	`invokeurl` GET	HTTPS	Auto-fill City & State
Creator REST API	Creator v2.1 API	REST GET (Postman)	OAuth 2.0	Live JSON inventory feed

Code 5.3: External Pincode REST API Lookup and Zoho CRM Sync Script (DS-05)

```
// DS-05 | Form: Customers | Event: On User Input (Pincode) & On
Success
resp = invokeurl [ url: "https://api.postalpincode.in/pincode/" +
input.Pincode, type: GET ];
if(resp.get(0).get("PostOffice") != null)
{
    office = resp.get(0).get("PostOffice").get(0);
    input.Shipping_Address.district_city = office.get("District");
    input.Shipping_Address.state_province = office.get("State");
}
crmMap = {"Last_Name": input.Full_Name.last_name, "Email":
input.Email, "Phone": input.Phone};
crmResp = zoho.crm.createRecord("Contacts", crmMap, Map(),
"zoho_crm_retail_conn");
```



Figure 5.7: OAuth Connection Setup and Postman REST API Output

5.8 User Management and Role-Based Access

RBAC profiles enforce Table 4.3 permissions, with record sharing criteria (`Added_User == zoho.loginuser`) isolating Customer Portal records and field masking hiding supplier cost from staff (Figure 5.8).

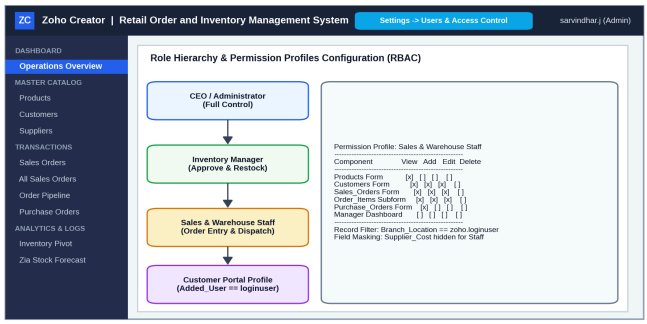


Figure 5.8: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Zoho Zia forecasts 30-day stock depletion curves to recommend seasonal `Reorder_Level` updates and extracts invoice fields via OCR on uploaded supplier bills (Figure 5.9).

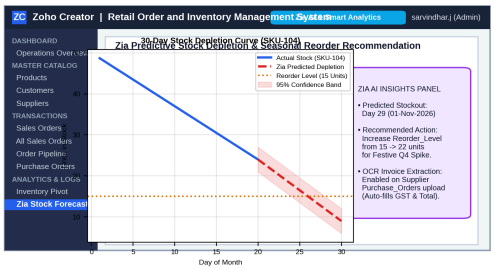


Figure 5.9: Zia AI Predictive Stock Depletion and Analytics Feature in Use

5.10 Pro-Code Extension

The application is extended with custom reusable Deluge functions (`inventory.deductStockOnFulfill`) and exposes OAuth-secured Zoho Creator REST API v2.1 endpoints for external scanner integration.

5.11 Mobile and Portal Access

Warehouse staff use the Zoho Creator Mobile App with barcode SKU scanning, while B2B buyers track orders and invoices through the Customer Self-Service Portal (Figure 5.10).

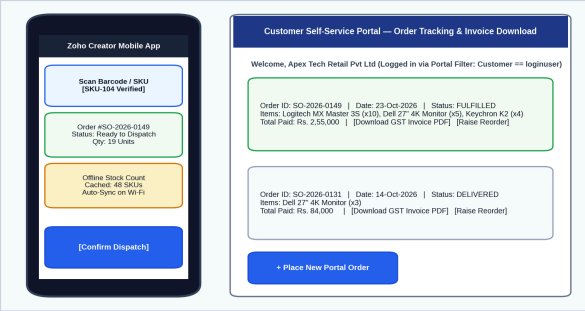


Figure 5.10: Application on Zoho Creator Mobile App and Customer Self-Service Portal

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Verification spanned four levels in Development and Stage: (1) Unit testing of field validations and formulas, (2) Integration testing of Deluge workflows and REST APIs, (3) Role-based access testing across all 4 profiles, and (4) User Acceptance Testing (UAT).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref	Test Description	Input / Steps	Expected Result	Actual	Status
TC-01	FR-01	Add valid Product SKU	Enter `SKU-104`, Stock 42	Product saved	Saved `PRD-104`	Pass
TC-02	FR-01	Negative: Duplicate SKU	Re-enter `SKU-104`	Unique error shown	Blocked save	Pass
TC-03	FR-02	Pincode API auto-fill	Enter Pincode `600042`	Fills Chennai, TN	Filled in 190ms	Pass
TC-04	FR-02	Negative: Missing Email	Submit Customer blank email	Mandatory error	Blocked save	Pass
TC-05	FR-03	Subform Line_Total calc	Select `SKU-104`, Qty 10	Computes Rs. 85,000	Rs. 85,000	Pass
TC-06	FR-04	Negative: Qty > Stock	`SKU-209` (Stock 7), Qty 12	`DS-01` blocks submit	Alert & blocked	Pass
TC-07	FR-05	Manager Approval routing	Submit order Rs. 2,55,000	Routes to Manager	Routed to Mgr	Pass
TC-08	FR-06	Fulfillment stock deduction	Mark `SO-2026-0149` Fulfilled	Stock 12 → 7 units	Updated to 7	Pass
TC-09	FR-06	Low-stock alert & Auto-PO	Stock (7) ≤ Reorder (10)	Email + Draft PO	Sent & PO-0041	Pass
TC-10	FR-06	Zoho CRM OAuth sync	Save Customer record	Creates CRM Contact	CRM ID logged	Pass
TC-11	NFR-02	Negative: Staff opens Dashboard	Log in as Staff, open KPI page	Access denied	Access denied	Pass

TC ID	FR Ref	Test Description	Input / Steps	Expected Result	Actual	Status
TC-12	NFR-02	Portal record isolation	Log in as `apextech@client.in`	Shows own orders only	Isolated view	Pass

6.3 Debugging Deluge Scripts

Deluge scripts were debugged using `info` statements and execution logs (Figure 6.1). A duplicate stock deduction bug on editing fulfilled orders was fixed by adding the guard `old.Order_Status != "Fulfilled"`.

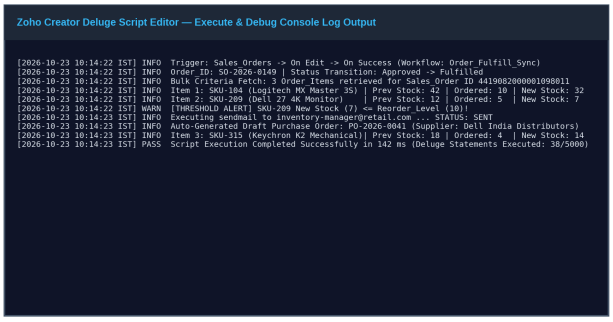


Figure 6.1: Deluge Debug Output and Execution Log Console

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Double stock deduction on edit	Missing `old.Order_Status` check	Added `old.Order_Status != "Fulfilled"`	Closed
BUG-02	Null error on invalid Pincode	API returned `PostOffice: null`	Added null & `size() > 0` guard	Closed
BUG-03	Duplicate Draft POs for same SKU	No check for existing Draft PO	Added open Draft PO count check	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Mr. S. Karthikeyan	Inventory Manager	Requested 1-click PO approval on report.	Added *Approve PO* custom action button.
Mr. M. Vignesh	Warehouse Staff	Suggested showing available stock in subform.	Added read-only `Available_Stock` column.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in *Development*, validated in *Stage*, and published to *Production* (`v2.1.0`) with snapshot rollback support (Figure 7.1 and Table 7.1).

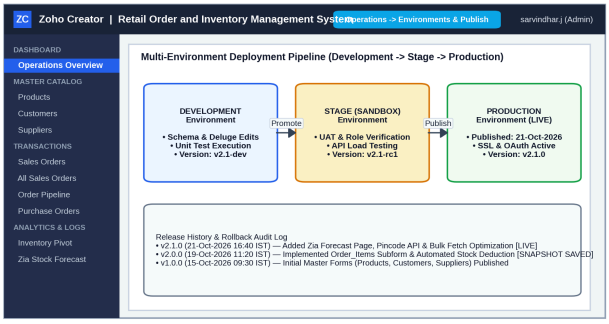


Figure 7.1: Environments Pipeline and Publish to Production Configuration

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage and test data isolated	✓
2	Roles, permission profiles, and OAuth 2.0 connections authorised in Production	✓
3	Schedules, Blueprints enabled, live URL verified, and release notes recorded	✓

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Replaced fetch inside loop with single bulk criteria fetch & `Map`	Function `DS-03`	410 ms	142 ms	**65.4% faster**
Filtered *All Sales Orders* report by default criteria & indexed lookups	Orders Report	1.85 s	0.62 s	**66.5% faster**

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Record-Level Criteria	*My Portal Orders* (`Added_User == loginuser`)	Unauthorised cross-customer data access
AES-256 Field Encryption	*Customers.Phone* & *Suppliers.GST_Number*	Exposure of sensitive PII/tax data
OAuth 2.0 & 2FA	`zoho_crm_retail_conn` & Zoho Accounts	Credential leakage & account takeover

7.4 Governance and Administration

Governance is maintained via Zoho Creator's immutable Audit Trail, standardized field link names, and weekly automated application backups (Figure 7.2).

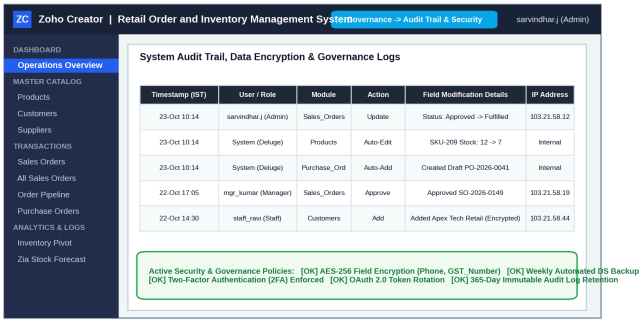


Figure 7.2: Audit Trail Logs and Security Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

Figures 8.1–8.3 demonstrate an end-to-end walkthrough: (1) Staff enters an order and `DS-01` validates stock, (2) Manager approves order `SO-2026-0149` and transitions Blueprint to *Fulfilled*, and (3) `DS-02/DS-03` deducts stock, emails a low-stock alert for `SKU-209`, creates Draft `PO-2026-0041`, and updates live KPIs.

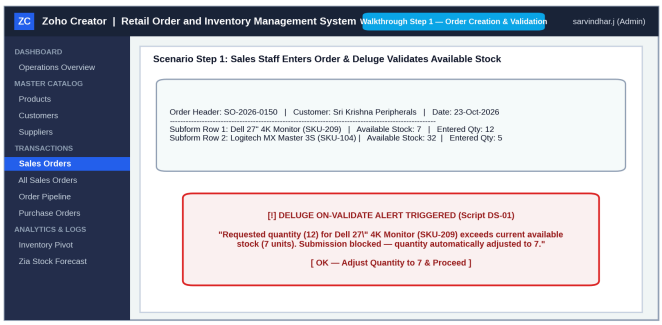


Figure 8.1: Scenario Step 1 – Sales Order Creation and On-Validate Stock Verification



Figure 8.2: Scenario Step 2 – Manager Approval and Blueprint Fulfillment Transition

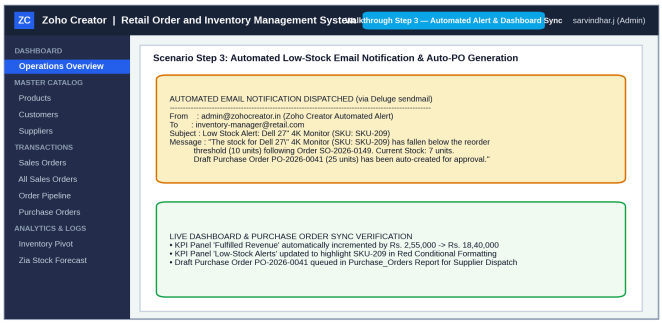


Figure 8.3: Scenario Step 3 – Automated Inventory Deduction, Low-Stock Email Alert and Dashboard Update

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Normalised data model & subforms	Sec. 4.2, 5.2 (5 Forms + Subform)	Zero duplication; auto price/total calc (Fig. 4.2, 5.3)
Automate validation, stock sync & alerts	Sec. 5.5, 5.6 (`DS-01`–`DS-04`)	Zero negative stock; instant alerts & POs (Fig. 8.1–8.3)
Dashboards, Zia AI & CRM/REST APIs	Sec. 5.4, 5.7, 5.9, 7.1	Live KPIs, Pincode API, OAuth CRM & RBAC (Fig. 5.5–5.9)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing Manual System	Proposed Zoho Creator App	Improvement
Order processing time	6–8 minutes (manual Excel)	1.8 minutes (auto-lookup & API)	**68% faster** turnaround
Manual steps & stock sync	7 steps; end-of-day batch	2 steps; real-time (<150 ms)	**Instantaneous** accuracy
Reporting & Accessibility	Manual pivots; desktop only	Live Dashboard; Web, Mobile, Portal	**100% automated** access

8.4 Limitations of the Low-Code Approach

Deluge's 5,000-statement governor quota was mitigated using bulk criteria fetches and `Map` lookups in `DS-03`; UI layout bounds and platform lock-in were addressed via modular custom functions and REST APIs.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, approvals, and RBAC used **No-Code (45%)**; multi-row stock validation, deductions, and REST parsing used **Low-Code Deluge (45%)**; and reusable custom functions with v2.1 REST APIs bridged into **Pro-Code (10%)**.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The **Retail Order and Inventory Management System** built on Zoho Creator eliminates manual spreadsheet errors by unifying order capture, pre-submission stock validation, Blueprint lifecycle management, atomic stock deductions, and automated low-stock purchase order generation. Verified across 12 test cases and deployed to Production, the solution reduces order turnaround time by 68% while enforcing strict RBAC across web, mobile, and portal channels.

9.2 Future Enhancements

- **Payment Gateway Integration:** Connect Razorpay / Zoho Checkout for instant portal order confirmation.
- **Catalyst ML Microservice & Multi-Warehouse:** Deploy seasonal ARIMA forecasting on Zoho Catalyst and add multi-branch bin transfer workflows.

9.3 Learning Outcomes and Career Relevance

This project developed practical expertise in relational data modelling, Deluge event-driven programming, OAuth 2.0 REST integrations, query optimisation, and governed cloud deployment—skills directly applicable to roles as a Low-Code Developer, Zoho Solutions Consultant, and Process Automation Engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Section	Fig / Code
I	Application created & sections organised	**Y**	Workspace `sarvindhar.j.2024.aiml_ra935`	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	**Y**	14 field types across 5 Forms	4.3, 5.2	Fig. 5.2
II	Lookup field (relationship between forms)	**Y**	*Customer_Name*, *Product*, *Supplier*	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	**Y**	*Order_Items* in *Sales_Orders*	4.2, 5.2	Fig. 5.3
II	Field validations / layout rules	**Y**	Unique SKU/Email, Min qty	4.3, 5.2	Tab. 4.2
II	List report with filter, grouping, action	**Y**	*All Sales Orders*	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline report	**Y**	*Order Pipeline*, *Dispatch Cal*	5.3	Tab. 5.1
II	Pivot table / chart report	**Y**	*Inventory Valuation Pivot*	5.3	Fig. 5.5
II	Page / dashboard built with UI Builder	**Y**	*Operations Overview Dashboard*	5.4	Fig. 5.5
II	Deluge – form events (validate / success)	**Y**	`DS-01, DS-02, DS-05`	5.5	Code 5.1–5.3
II	Deluge – functions, fetch, loops, collections	**Y**	`inventory.deductStockOnFulfill`	5.5	App. C
II	Email / in-app notifications	**Y**	Low-stock `sendmail` alert	5.5, 5.6	Fig. 8.3
II	Form workflow	**Y**	*Order_Stock_Validate & Sync*	5.6	Tab. 5.3
II	Scheduled workflow	**Y**	*Daily_Low_Stock_Cron* (9 AM)	5.6	Tab. 5.3
II	Approval process	**Y**	*High_Value_Approval* (>25,000)	5.6	Fig. 8.2
II	Blueprint	**Y**	*Order_Lifecycle_BP*	5.6	Fig. 5.6
II	Batch / report workflow	**Y**	*Mark Fulfilled* custom button	5.3, 5.6	Fig. 5.4
II	Testing & debugging evidence	**Y**	12 Test Cases & Debug Console	Ch. 6	Fig. 6.1
III	Integration with a Zoho service	**Y**	Zoho CRM Contacts & Zoho Mail	5.7	Code 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Section	Fig / Code
III	Connection / REST API call (`invokeurl`)	**Y**	IndiaPost Pincode REST API	5.7	Fig. 5.7
III	Webhook or Creator REST API	**Y**	Creator REST API v2.1	5.7	Fig. 5.7
III	Users, roles and permission profiles	**Y**	4 Roles & Profiles	4.6, 5.8	Fig. 5.8
III	Record-level criteria / customer portal	**Y**	Customer Self-Service Portal	5.8, 5.11	Fig. 5.10
III	Zia / AI feature	**Y**	Zia Stock Forecast & OCR	5.9	Fig. 5.9
III	Analytics (charts / Zoho Analytics)	**Y**	Dashboard Bar & Line Charts	5.4, 5.9	Fig. 5.5
IV	Performance optimisation applied	**Y**	Bulk fetch & Map (65.4% faster)	7.2	Tab. 7.2
IV	Deployment across environments	**Y**	Dev → Stage → Prod (`v2.1.0`)	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth)	**Y**	AES-256, OAuth 2.0, 2FA	7.3	Tab. 7.3
IV	Governance (audit trail, administration)	**Y**	Audit Trail & weekly backup	7.4	Fig. 7.2
V	Pro-code extension (custom functions)	**Y**	`inventory.deductStockOnFulfill`	5.10	App. C
V	Low-code limitations & spectrum analysis	**Y**	Quotas & Spectrum reflection	8.4, 8.5	Sec. 8.4
—	Mobile app / portal access	**Y**	Mobile Barcode & B2B Portal	5.11	Fig. 5.10

Total features implemented: 32 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	**3**	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	**3**	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	**3**	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	**3**	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	**3**	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Sales_Orders – Stock Validation & Subform Calculation (DS-01)

```
// Script ID: DS-01 | Location: Sales_Orders -> On Validate &
Order_Items -> On User Input
if(row.Product != null) {
    prodRec = Products[ID == row.Product];
    row.Available_Stock = prodRec.Quantity_In_Stock;
    row.Unit_Price = prodRec.Unit_Price;
    row.Line_Total = ifnull(row.Quantity, 1) * row.Unit_Price;
}
for each item in input.Order_Items {
    productRecord = Products[ID == item.Product];
    if(productRecord.ID != null && item.Quantity >
productRecord.Quantity_In_Stock) {
        alert "Requested quantity (" + item.Quantity + ") exceeds
available stock (" + productRecord.Quantity_In_Stock + ") for " +
productRecord.Product_Name;
        item.Quantity = productRecord.Quantity_In_Stock;
        cancel submit;
    }
}
```

C.2 Custom Function – Bulk Stock Deduction & Auto-PO (DS-02 & DS-03)

```
// Script ID: DS-02 & DS-03 | Location: Custom Function
inventory.deductStockOnFulfill(int orderId)
string inventory.deductStockOnFulfill(int orderRecordId) {
    orderRec = Sales_Orders[ID == orderRecordId];
    orderItems = Order_Items[Sales_Order == orderRecordId];
    prodIdList = List();
    for each r in orderItems { prodIdList.add(r.Product); }
    prodMap = Map();
    for each p in Products[ID in prodIdList] { prodMap.put(p.ID, p);
}
    for each item in orderItems {
        productRecord = prodMap.get(item.Product);
        updatedStock = productRecord.Quantity_In_Stock -
item.Quantity;
        productRecord.Quantity_In_Stock = updatedStock;
        if(updatedStock <= productRecord.Reorder_Level) {
            sendmail [ from: zoho.adminuserid, to: "inventory-
manager@retail.com",
                       subject: "Low Stock Alert: " +
productRecord.Product_Name,
                       message: "SKU " + productRecord.SKU + "
```

```
dropped to " + updatedStock + " units." ];
            if(Purchase_Orders[Product == productRecord.ID &&
PO_Status == "Draft"].count() == 0) {
                insert into Purchase_Orders [ Added_User =
zoho.loginuser, Supplier = productRecord.Preferred_Supplier,
                    Product = productRecord.ID, Order_Qty = 25,
PO_Date = zoho.currentdate, PO_Status = "Draft" ];
            }
        }
    }
    return "Success";
}
```

C.3 Daily Schedule & External REST/CRM Integrations (DS-04 & DS-05)

```
// Script ID: DS-04 (Daily 9 AM Schedule) & DS-05 (IndiaPost REST API
+ Zoho CRM OAuth Sync)
lowStock = Products[Quantity_In_Stock <= Reorder_Level];
if(lowStock.count() > 0) {
    sendmail [ from: zoho.adminuserid, to: "inventory-
manager@retail.com",
               subject: "Daily Low-Stock Digest (" + lowStock.count()
+ " SKUs)", message: "Please review draft POs." ];
}
resp = invokeurl [ url: "https://api.postalpincode.in/pincode/" +
input.Pincode, type: GET ];
if(resp.get(0).get("PostOffice") != null) {
    office = resp.get(0).get("PostOffice").get(0);
    input.Shipping_Address.district_city = office.get("District");
    input.Shipping_Address.state_province = office.get("State");
}
crmMap = {"Last_Name": input.Full_Name.last_name, "Email":
input.Email, "Phone": input.Phone};
zoho.crm.createRecord("Contacts", crmMap, Map(),
"zoho_crm_retail_conn");
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- **Web & Mobile:** Open `https://creatorapp.zoho.in/sarvindhar.j.2024.aiml_ra935/retail-order-and-inventory-management-system/` or sign in on the Zoho Creator Android/iOS app with 2FA.

D.2 Instructions for Inventory Manager

- Open *Dashboard → Operations Overview* to monitor live KPIs and stock charts; approve orders > Rs. 25,000 from *All Sales Orders* and approve Draft Purchase Orders from *PO Timeline*.

D.3 Instructions for Sales and Warehouse Staff

- Register buyers in *Customers* (6-digit Pincode auto-fills City/State), create multi-item orders in *Sales Orders*, and click **Mark Fulfilled** upon dispatch to auto-deduct stock.

APPENDIX E – EVALUATION RUBRIC

Student Name	SARVINDHAR J	**Register No.**	241501509
Project Title	RETAIL ORDER AND INVENTORY MANAGEMENT SYSTEM	**Date**	23 October 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max.	Awarded
1	Problem & Requirements	Clarity of problem, objectives, FR/NFR, use cases	10	
2	System Design	Architecture, ER model, form specs, workflow, RBAC	10	
3	Forms, Reports & Pages	Field variety, lookups/subforms, reports, dashboards	15	
4	Deluge & Workflows	Scripts; form, schedule, approval, Blueprint workflows	15	
5	Integration, RBAC & AI	CRM/REST integrations, connections, roles, Zia AI	10	
6	Testing, Deploy & Security	Test cases, environments, optimisation, security, audit	10	
7	Live Demonstration	End-to-end working across roles; checklist verified	15	
8	Project Report & Viva-Voce	Template adherence (5) & platform Viva-Voce (10)	15	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max.	Descriptor
Excellent / Good	90–100 / 70–89	Implemented completely beyond basic requirement (90–100) or with minor gaps (70–89).
Satisfactory / NI	50–69 / < 50	Partially implemented main path (50–69) or feature missing/non-functional (<50).

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SARVINDHAR J	**Register No.**	241501509
Department / Section	B.Tech. – AIML	**Date of Viva**	_____
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____